

Combined Plaque Radiotherapy and Transpupillary Thermotherapy for Choroidal Melanoma

Tumor Control and Treatment Complications in 270 Consecutive Patients

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Objective: To evaluate tumor control and treatment complications following plaque radiotherapy combined with transpupillary thermotherapy for choroidal melanoma.

Design: Prospective noncomparative interventional case series.

Intervention: All patients received treatment for choroidal melanoma using plaque radiotherapy followed by 3 sessions of transpupillary thermotherapy provided at plaque removal and at 4-month intervals.

Participants: Two hundred seventy patients with newly diagnosed choroidal melanoma.

Main Outcome Measures: The 2 main outcome measures included local tumor recurrence and treatment-related complications. The clinical data regarding patient features, tumor features, radiotherapy and thermotherapy parameters were analyzed for their effect on the 2 main outcomes using Cox proportional hazards regression models.

Results: Prior to treatment, the median base of the tumor was 11 mm (range, 4-21 mm) and the median thickness was 4 mm (range, 2-9 mm). Most tumors were

located in the posterior pole with a median proximity of 2 mm to the foveola and 2 mm to the optic disc. The median radiotherapy dose to the tumor apex was 9000 rad. Transpupillary thermotherapy was applied in 3 sessions at 4-month intervals for a median of 700 mW. The tumor decreased in thickness to a median of 2.3 mm by 1 year and 2.1 mm by 2 years' follow-up with stable findings thereafter. Using Kaplan-Meier estimates, tumor recurrence was 2% at 2 years and 3% at 5 years. Risk factors for tumor recurrence included macular location of the tumor epicenter ($P = .03$), diffuse tumor configuration ($P = .005$), and tumor margin extending underneath the foveola ($P = .001$). Using Kaplan-Meier estimates, treatment-related complications at 5 years included maculopathy in 18% of the participants, papillopathy in 38%, macular retinal vascular obstruction in 18%, vitreous hemorrhage in 18%, rhegmatogenous retinal detachment in 2%, cataract in 6%, and neovascular glaucoma in 7%. Enucleation for radiation complications was necessary in 3 cases (1%).

Conclusion: Plaque radiotherapy combined with transpupillary thermotherapy provides excellent local tumor control with only 3% recurrence at 5 years' follow-up.

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PLAQUE RADIOTHERAPY has assumed a major role in the management of posterior uveal melanoma.¹⁻¹² Plaque radiotherapy is a form of brachytherapy in which a radiation implant is designed and placed surgically on the sclera directly over an intraocular tumor to provide maximal radiation dose to the tumor, while minimizing the dose to normal tissues.⁴ The method of brachytherapy was first introduced for uveal melanoma in the late 1920s using radon seed insertion directly into intraocular melanoma.⁵ Later revisions on the technique allowed transscleral delivery of radiotherapy via a tagged cobalt 60 implant or plaque sutured onto the episclera.⁶ Since

then, several radioisotopes have been used, including ruthenium 106, iridium 192, iodine 125, and palladium 103.³ Presently, ¹²⁵I is the most commonly used isotope for plaque radiotherapy of choroidal melanoma.

Plaque radiotherapy is designed as an eye-conserving treatment for intraocular cancers and it is generally tolerated by the ocular structures. A review of 1300 patients with uveal melanoma treated with this method revealed local ocular irradiation complications such as nonproliferative retinopathy in 42% of those treated and proliferative retinopathy in 8% of those treated at 5 years' follow-up.⁷ These complications, combined with the presence of an intraocular mass and frequent associ-

PATIENTS AND METHODS

PATIENT DATA

Data from all patients with the diagnosis of uveal melanoma treated with ^{125}I plaque radiotherapy combined with subsequent planned transpupillary thermotherapy on the Ocular Oncology Service, Wills Eye Hospital, Philadelphia, Pa, between January 1, 1995, and January 1, 2000, were prospectively collected. Data included features of the patient and tumor as well as irradiation parameters. Clinical data were then analyzed for the outcomes of tumor recurrence and treatment-related complications.

Data included patient features at initial examination such as age, race (African American, Hispanic, Asian, or *white*), sex (female or *male*), medical problems (*none*, diabetes mellitus, hypertension, or hypercholesterolemia), and previous or present use of chemotherapy. The best-corrected Snellen visual acuity was measured at 20 ft (20/20, 20/25, 20/30, 20/40, 20/50, 20/60, 20/70, 20/80, 20/100, 20/200, 20/400, counting fingers, hand motions, light perception, and no light perception). Reference categories that were used in subsequent statistical analyses are italicized.

The tumor data included anatomical location (ciliary body, ciliochoroidal, or *choroidal*), meridian location of tumor epicenter (superior, superotemporal, temporal, inferotemporal, inferior, *inferonasal*, nasal, superonasal, or macula), proximity to optic nerve and foveola (in millimeters), location of anterior and posterior tumor margins (iris, ora serrata, between ora serrata and equator, or *posterior to equator*), largest basal dimension (based on ophthalmoscopy in millimeters), largest thickness (based on ultrasonography in millimeters), shape (*dome*, mushroom, diffuse, or plateau), pigmentation (amelanotic or *melanotic*), and subretinal fluid (*absent* or present).

The ^{125}I radiation plaque data included plaque shape (*round*, notched, curvilinear, or rectangular), plaque size, hours of radiation exposure, and total radiation dose (in rad) and radiation dose rate (rad per hour) to the tumor apex, tumor base, optic disc, foveola, and lens. The target apex dose was 8000 to 10000 rad with radiation design and parameters as previously published.^{1,3} Parameters regarding adjuvant treatment with transpupillary thermotherapy were recorded regarding power setting (milliwatts), spot size (1.2 mm, 2.0 mm, or 3.0 mm), delivery method (indirect ophthalmoscope

adapter, slitlamp adapter), number of treatment spots, total duration of treatment (in minutes), and tumor uptake (*none*, light, or heavy). The first session of thermotherapy was applied at the time of plaque removal in the operating room and the second and third planned sessions at 4-month intervals were provided in the office setting at the slitlamp unit using previously published techniques.^{14,15}

Follow-up examinations were generally made at 4-month intervals up to 5 years and 6- to 12-month intervals thereafter. Data regarding tumor recurrence and treatment-related complications were recorded. Tumor recurrence was defined as documented tumor growth in thickness of at least 0.4 mm or base of at least 0.2 mm as detected by ophthalmoscopy, fundus photography, or ultrasonography. Treatment-related complications including radiotherapy- or thermotherapy-induced retinopathy, maculopathy, papillopathy, macular and extramacular retinal vascular obstruction, surface wrinkling retinopathy, rhegmatogenous retinal detachment, vitreous hemorrhage, cataract, and neovascular glaucoma were recorded. Radiation retinopathy and maculopathy were defined as retinal capillary bed changes (nonperfusion, dilation, microaneurysm, or hemorrhage), retinal exudation, retinal edema, nerve fiber layer infarction, or vascular sheathing.⁷ Radiation papillopathy was defined as peripapillary exudation, hemorrhage, or edema, as well as optic disc edema.

STATISTICAL ANALYSIS

The 2 main outcomes in this article were local tumor recurrence and treatment-related complications. The effect of individual clinical variables on the development of tumor recurrence was analyzed by a series of univariate Cox proportional hazards regressions.¹⁷ The correlation among the variables was determined by using Pearson product moment correlations. All variables were analyzed as discrete variables except for patient age, intraocular pressure, tumor base, tumor thickness, proximity to the optic disc, proximity to the foveola, percentage of tumor overhanging the optic disc, radiation dose, radiation rate, thermotherapy power, and duration of thermotherapy, which were analyzed as continuous variables and by grouping into categories. Multivariable analysis was not feasible owing to the few recurrences. The Kaplan-Meier method was used to estimate local tumor recurrence and treatment-related complications as a function of time.¹⁸

ated serous retinal detachment, led to ultimate poor visual acuity (20/200 or worse) in 34% of the patients at 5 years' and 68% of the patients at 10 years' follow-up.⁸ Thus, despite successful treatment of the uveal melanoma and preservation of the eye, visual acuity is reduced in most patients.

An important goal of plaque radiotherapy for uveal melanoma is to achieve local control of the melanoma.⁹⁻¹² Karlsson et al⁹ found the 5-year local relapse following ^{60}Co plaque radiotherapy to be 12%. Lommatzsch et al¹² reported a higher cumulative relapse rate of 37% at 15 years following ^{106}Ru plaque radiotherapy. Wilson and Hungerford¹¹ analyzed a large group of patients with choroidal melanoma treated with various radiotherapy methods and found that local tumor recurrence at 5 years' follow-up was 4% with ^{125}I plaque radiotherapy, 11% with ^{106}Ru plaque

radiotherapy, and 5% with proton beam radiotherapy. Their comparative analysis of the 3 treatment methods showed that ^{125}I and proton beam radiotherapy provided greater local tumor control than ^{106}Ru .

We have been using plaque radiotherapy for almost 30 years in the management of posterior uveal melanoma. Although we used ^{60}Co , ^{192}Ir , and ^{106}Ru radioisotopes in the first few years, we currently use ^{125}I almost exclusively. More recently we have used a combination of plaque radiotherapy combined with argon laser photocoagulation or transpupillary thermotherapy to secure better local tumor control, especially with those tumors located near the optic disc and foveola. Transpupillary thermotherapy using infrared laser is more penetrating and destructive to melanoma than argon laser photocoagulation and is our preferred adjunctive treat-

Table 1. Plaque Radiotherapy Combined With Transpupillary Thermotherapy for Choroidal Melanoma in 270 Cases: Patient Demographics*

Variable	No. (%) of Patients
Age, y	
Median (mean, range)	57 (58, 15-93)
Sex	
Male	137 (51)
Female	133 (49)
Race	
White	268 (99)
African American	2 (1)
Medical history*	
Absent	190 (70)
Hypertension	62 (23)
Diabetes mellitus	24 (9)
Hypercholesterolemia	2 (<1)
Chemotherapy	0

*Some patients had more than 1 condition.

ment.¹³⁻¹⁶ In this article, we analyze local tumor control and complications using custom-designed ¹²⁵I plaque radiotherapy combined with transpupillary thermotherapy.

RESULTS

The general information regarding patient demographics is listed in **Table 1**. The visual acuity at presentation was 20/20 to 20/50 in 206 eyes (76%), 20/60 to 20/100 in 36 eyes (13%), and 20/200 to no light perception in 28 eyes (10%). Data regarding tumor characteristics are listed in **Table 2**. There were 200 melanomas (74%) that were 3 mm or less from the optic disc and 95 melanomas (35%) that were touching the optic disc. There were 205 melanomas (76%) that were 3 mm or less from the foveola and 65 melanomas (24%) that were underneath the foveola. The radiotherapy and thermotherapy parameters are listed in **Table 3**.

The patients were followed up for a median of 29 months (mean, 32 months; range, 5-79 months) from the initiation of plaque radiotherapy and thermotherapy. The interquartile range (25%-75%) was 17 to 43 months. The final visual acuity was 20/20 to 20/50 in 101 eyes (37%), 20/60 to 20/100 in 44 eyes (16%), and 20/200 to no light perception (including enucleation) in 125 eyes (46%). The 2 outcomes of local tumor recurrence and treatment-related complications are summarized in **Table 4**. Only 5 in the entire group of 270 patients developed local tumor recurrence. The recurrence was predominantly located on the posterior margin in all 5 cases. The recurrence was detected at a median interval of 24 months (mean, 21 months; range, 8-27 months) following plaque radiotherapy and at the time of detection, the recurrence was less than 1 mm in base in 1 case and 2 mm or greater in base in 4 cases. Treatment of the recurrence involved further thermotherapy in 1 case and enucleation in 4 cases. Using Kaplan-Meier estimates, local tumor recurrence was less than 1% at 1 year's, 2% at 2 years', 3% at 3 years', and 3% at 5 years' follow-up (Table 4)

Table 2. Plaque Radiotherapy Combined With Transpupillary Thermotherapy for Choroidal Melanoma in 270 Cases: Tumor Characteristics

Features	No. (%) of Patients
Tumor location	
Choroidal	265 (98)
Ciliochoroidal	5 (2)
Tumor meridian	
Superior	22 (8)
Superotemporal	15 (6)
Temporal	11 (4)
Inferotemporal	16 (6)
Inferior	14 (5)
Inferonasal	27 (10)
Nasal	24 (9)
Superonasal	21 (8)
Macular	120 (44)
Proximity to optic disc, mm	
Median (mean, range)	2 (2, 0-9)
Proximity to foveola, mm	
Median (mean, range)	2 (2, 0-10)
Tumor base, mm	
Median (mean, range)	11 (11, 4-21)
Tumor thickness, mm	
Median (mean, range)	4 (4, 2-9)
Tumor shape	
Dome	170 (63)
Diffuse	38 (14)
Mushroom	32 (12)
Plateau	30 (11)
Associated subretinal fluid	
Present	250 (93)
Absent	20 (7)
Tumor color	
Melanotic	237 (88)
Amelanotic	33 (12)

(**Figure 1**). By univariable analysis, factors at the initial examination and treatment predictive of local tumor recurrence are noted in **Table 5** and included tumor epicenter location in the macula ($P=.03$), diffuse tumor shape ($P=.005$), and tumor extending underneath the foveola ($P=.001$). Multivariable analysis was infeasible owing to the few recurrences. Of the 270 patients, there were 95 tumors (35%) that touched the optic disc and were classified as juxtapapillary choroidal melanoma. Of these 95 juxtapapillary tumors treated with plaque radiotherapy combined with transpupillary thermotherapy, 2.5% (95% confidence interval [CI]=0.0%-3.2%) recurred by 5 years' follow-up. Thirty-three of the 95 juxtapapillary tumors hung over the optic disc. When overhanging the optic disc, the tumor obscured a median of 50% (mean, 47%; range, 2%-100%) of the optic disc. Only 5% of the patients with tumor overhanging the optic disc showed tumor recurrence following treatment at 5 years. Of the 270 patients, there were 65 patients (24%) with tumors underneath the foveola and 5% showed recurrence at 5 years' follow-up.

Using Kaplan-Meier estimates, the main treatment-related complications are summarized in Table 4 and **Figure 2**. The most common complications at 5 years following treatment included retinopathy in 39% and pap-

Table 3. Plaque Radiotherapy Combined With Transpupillary Thermotherapy for Choroidal Melanoma in 270 Cases: Treatment Parameters

Variables	Median (Range)
Plaque parameters	
Radiation duration, h	96 (70-139)
Radiation dose, rad	
Apex	9000 (5500-12 400)
Base	25 450 (10 800-49 000)
Optic disc	5639 (1085-66 686)
Foveola	6244 (921-40 913)
Radiation rate, rad/h	
Apex	93 (7-318)
Base	267 (118-534)
Optic disc	60 (12-367)
Foveola	65 (10-447)
Transpupillary thermotherapy parameters	
Session 1	
Power, mW	700 (300-1600)
Duration, min	7 (1-32)
Spot size, mm	1.2 (1.2)
Session 2	
Power, mW	700 (400-1400)
Duration, min	9 (4-25)
Spot size, mm	3 (1.2-3)
Session 3	
Power, mW	800 (300-1400)
Duration, min	8 (2-23)
Spot size, mm	3 (1.2-3)

Table 4. Plaque Radiotherapy Combined With Transpupillary Thermotherapy for Choroidal Melanoma in 270 Cases: Summary of Outcomes (Tumor Recurrence and Main Treatment-Related Complications)*

Outcomes	% at 2 Years Using Kaplan-Meier Estimates	% at 5 Years Using Kaplan-Meier Estimates
Local tumor recurrence	2	3
Treatment-related complications		
Retinopathy	5	39
Maculopathy	4	18
Papillopathy	4	38
BRAO/BRVO		
Extramacular	9	15
Macular	8	18
Surface wrinkling retinopathy	8	8
Rhegmatogenous retinal detachment	2	2
Vitreous hemorrhage	1	18
Cataract	2	6
Neovascular glaucoma	<1	7

*BRAO/BRVO indicates branch retinal artery obstruction–branch retinal vein obstruction.

illopathy in 38%. Those patients who developed papillopathy generally had tumors within 3 mm of the optic disc ($P<.001$; relative risk, 1.40; 95% CI, 1.15-1.70).

Melanoma metastasis developed in 13 patients at 10 years' follow-up. Kaplan-Meier estimates of the rate of metastasis was 3% at 2 years and 12% at 5 years (Figure 2K). Twelve of the 13 patients showed no local tumor recurrence and 1 patient had demonstrated previous local recurrence.

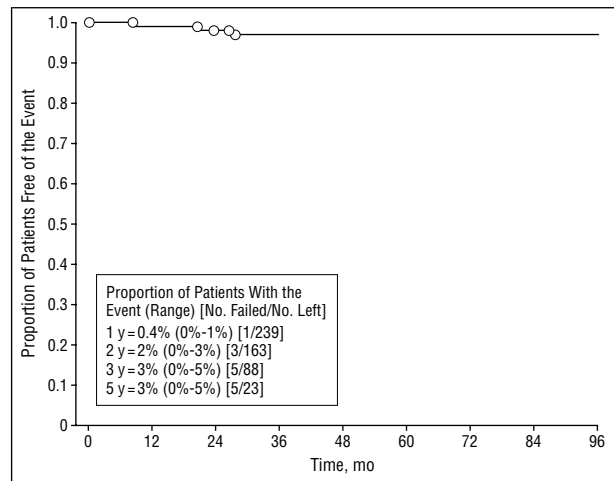


Figure 1. Plaque radiotherapy combined with transpupillary thermotherapy for choroidal melanoma in 270 cases: Kaplan-Meier estimates of patients free of local tumor recurrence.

COMMENT

Several reports have addressed the issue of local tumor recurrence following radiotherapy of uveal melanoma. In 1989, Karlsson et al⁹ reported 12% local tumor recurrence at 5 years following ⁶⁰Co plaque radiotherapy in 277 patients. They found that the predictors for recurrence included increasing largest linear tumor dimension and increasing nearness of the tumor margin to the optic disc. Thus, juxtapapillary melanoma with large dimension was at greatest risk for recurrence following ⁶⁰Co plaque radiotherapy. Later, Lommatzsch et al¹² published results of long-term tumor control with ¹⁰⁶Ru plaque radiotherapy in 141 eyes with choroidal melanoma managed over 12 years, with a median follow-up of 17 years. They found cumulative 15-year local tumor recurrence in 37% and confirmed that greater tumor diameter was the main factor associated with recurrence. In 1999, Wilson and Hungerford¹¹ reported their experience with plaque radiotherapy using either ¹²⁵I or ¹⁰⁶Ru applicators as well as proton beam radiotherapy. They found tumor recurrence at 5 years was 4% with ¹²⁵I, 11% with ¹⁰⁶Ru, and 5% with proton beam radiotherapy. They indicated that ¹²⁵I provided statistically significant better tumor control than ¹⁰⁶Ru. Proton beam radiotherapy showed no significant improvement in control over ¹⁰⁶Ru according to their analysis.

The issue of tumor control is important for conservation of the eye and visual acuity. More importantly, local tumor recurrence influences the overall patient survival. Karlsson et al⁹ found that patients who displayed local tumor recurrence within the eye were at greater risk for distant, life-threatening tumor metastasis. Of those with local recurrence, 42% developed metastasis at 5 years compared with 18% metastasis in those patients without local recurrence. These results were further confirmed by Vrabec et al,¹⁹ who evaluated survival of 62 patients with local relapse compared with a matched group. They found that the local relapse group showed metastasis in 42% compared with 13% for those without local relapse. They speculated that local tumor relapse indicated greater malignant potential of uveal melanoma.

Table 5. Plaque Radiotherapy Combined With Transpupillary Thermotherapy for Choroidal Melanoma in 270 Cases: Factors at Initial Presentation Predictive of Tumor Recurrence

Univariable Analysis*	No Tumor Recurrence, No. (%) of Cases (n = 265)	Tumor Recurrence, No. (%) of Cases (n = 5)	P Value	Relative Risk	95% Confidence Interval
Tumor epicenter location (macula vs other†‡)	114 (43)	5 (100)	.03	...	...
Tumor shape (diffuse vs other†)	34 (13)	4 (80)	.005	22.80	2.55-204.11
Tumor proximity to foveola (0 mm vs >0 mm†)	54 (20)	4 (80)	.001	15.05	1.69-134.67

*Multivariable analysis was infeasible owing to the few recurrences.

†Reference variable.

‡Cox proportional hazards regression, score test used for P value; relative risk and 95% confidence interval are not estimable owing to the presence of zero cell.

The technique of combining plaque radiotherapy with thermotherapy was conceived with the idea of minimizing local recurrence and improving overall survival. In this study, we found improved local tumor control with 97% at 5 years' follow-up using ^{125}I plaque radiotherapy combined with thermotherapy (Figure 1), compared with previous observations of 88% local tumor control using ^{60}Co applicators. Such control could be owing to many factors including refined accuracy with localization of the tumor and placement of the plaque, advances in the adaptation of plaque design for unusual locations such as notched-plaque design for tumors at or overhanging the optic disc (**Figure 3** and **Figure 4**), improvements in the ability to customize a radioactive plaque for each patient using an array of ^{125}I seeds rather than the noncustomized approach with the fixed, standard ^{60}Co applicators, and perhaps the addition of adjuvant transpupillary thermotherapy.²⁻⁴ Thus, the combination of these factors may have allowed for exceptional local control and likewise preservation of the eye in 99% cases.

Plaque radiotherapy can be used for uveal melanoma at almost any site within the eye.²⁰⁻²² In particular, plaque radiotherapy has been adapted for choroidal melanoma in technically difficult sites such as juxtapapillary and macular melanomas. We previously reported local tumor recurrence in 19% of plaque-irradiated juxtapapillary choroidal melanomas²³ and 9% of plaque-irradiated macular choroidal melanomas²⁴ at 5 years' follow-up. However, when those patients were treated, notched plaques and thermotherapy were unavailable; thus they were treated with a standard round plaque. In contrast, in the present study, custom notched plaques and adjuvant thermotherapy were used in all 95 patients with juxtapapillary choroidal melanoma (Figures 3 and 4) and we found improved tumor control, with only 2.5% of patients showing local tumor relapse at 5 years. Of the 33 patients in our group with more advanced juxtapapillary choroidal melanoma overhanging the optic disc (Figure 4), recurrence was found in only 5% at 5 years. In these more advanced cases, a special plaque design with a deeply notched posterior aspect and radioactive seeds along the margin of the notch provides adequate radiation dose and rate to the entire tumor, especially the prepapillary portion of the melanoma. For macular choroidal melanoma, there were 65 patients with tumor extending underneath the foveola in our present

group, and 5% showed recurrence at 5 years, which is an improvement over the previously published recurrence rate of 9%.²⁴

Tumor control with ^{125}I plaque radiotherapy is equivalent to that found with proton beam radiotherapy as reported by Wilson and Hungerford¹¹ and now implied in our results using the addition of supplemental thermotherapy. Gragoudas et al²⁵ found 3% tumor recurrence at 5 years following proton beam radiotherapy of uveal melanoma. In their report, they included only recurrences that measured 1 mm or larger and excluded those smaller than 1 mm. In our study, all clinically visible recurrences were included, regardless of size, even as small as 0.2 mm increase in base. Even with the inclusion of all recurrences, our results are comparable with only 3% relapse at 5 years. If we only included tumor recurrences of 1 mm or larger, our local relapse rate would have been even less. This information suggests that plaque radiotherapy and proton beam radiotherapy performed at experienced centers for ocular oncology provide excellent local tumor control of choroidal melanoma.

It is important to understand the complications of radiotherapy and thermotherapy to the eye (Figures 2A-J) (Table 4).^{7,15,16,24,26} Radiation retinopathy following plaque radiotherapy has been found in 42% of patients with the development of proliferative retinopathy in 8% at 5 years.⁷ Factors predictive of retinopathy in patients with choroidal melanoma treated with ^{125}I applicators included tumor thickness over 5 mm and elevated tumor base dose rate exceeding 260 rad/h.⁷ Visual acuity loss is most profound in those patients with thicker tumors near the optic disc and foveola.⁸ The vision-threatening complications are greater for patients with tumors at posterior locations such as those near the optic disc and those in the macula. At 5 years' follow-up of plaque radiotherapy for juxtapapillary choroidal melanoma, radiation retinopathy was found in 95% and papillopathy in 58%, leading to visual loss of at least 3 Snellen lines in 70%.²³ Similarly, patients with macular choroidal melanoma underlying the foveola showed vision impairing radiation maculopathy in 40% and papillopathy in 13% at 5 years following plaque radiotherapy.²⁴ Our group of 270 patients displayed juxtapapillary tumors in 95 cases (35%) and subfoveal tumors in 65 cases (24%), explaining the bothersome posterior segment complications of radiation maculopathy in 18% and papillopathy in 38% at 5 years. Thermotherapy can also lead to visually com-

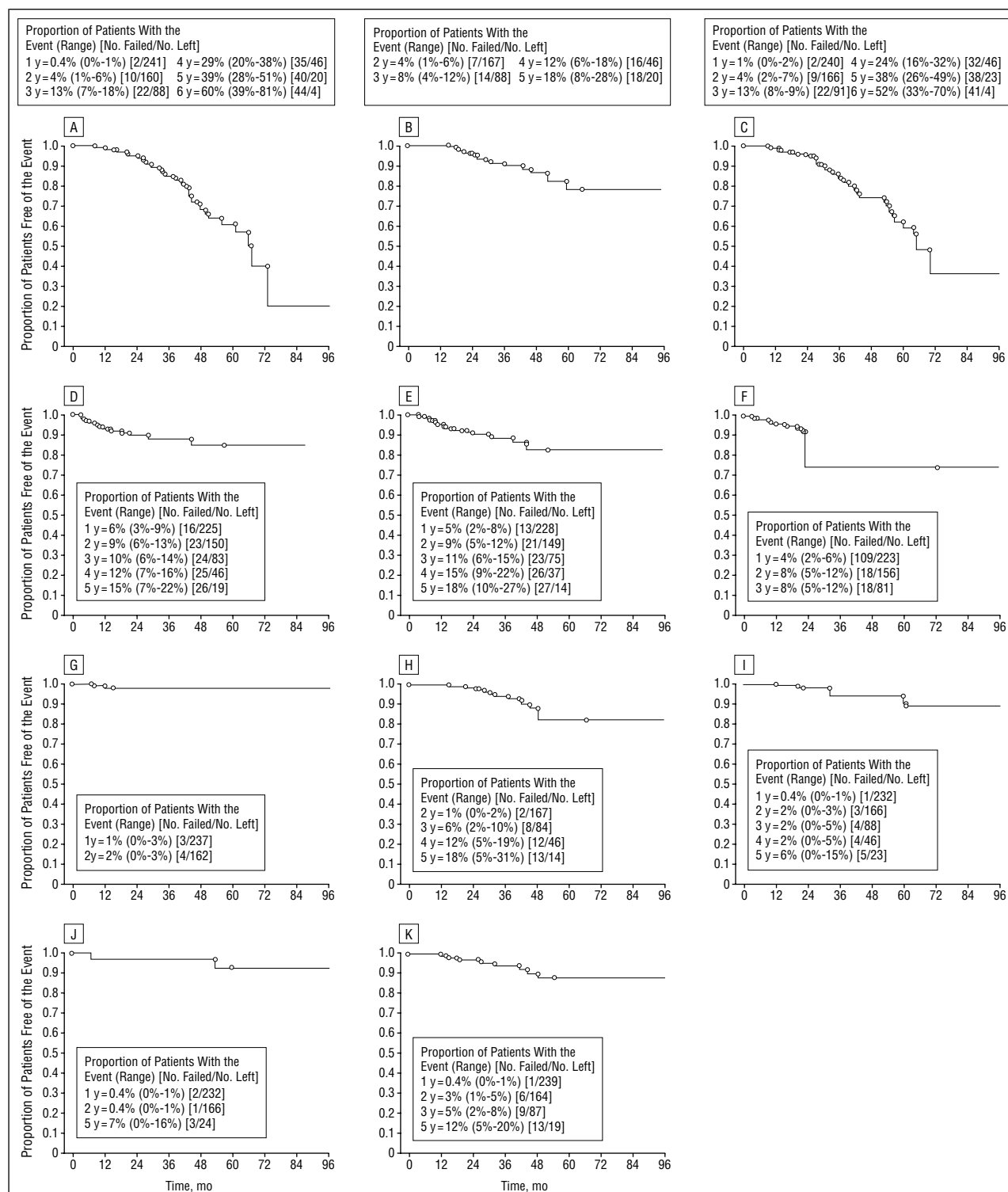


Figure 2. Plaque radiotherapy combined with transpupillary thermotherapy for choroidal melanoma in 270 cases: Kaplan-Meier estimates of patients free of radiation retinopathy (A), radiation maculopathy (B), radiation papillopathy (C), extramacular branch retinal vein obstruction or branch retinal artery obstruction (D), macular branch retinal vein obstruction or branch retinal artery obstruction (E), surface wrinkling retinopathy (F), rhegmatogenous retinal detachment (G), vitreous hemorrhage (H), radiation cataract (I), neovascular glaucoma (J), and metastasis (K).

promising complications such as macular retinal vascular obstruction and surface wrinkling retinopathy, found in 18% and 8% of our patients, respectively, at 5 years.^{15,16}

Plaque radiotherapy for choroidal melanoma provides equivalent local tumor control and similar posterior segment complications, but substantially less ante-

rior segment complications than charged particle irradiation using proton beam or helium ion.^{10,11} Char et al¹⁰ evaluated 184 patients with choroidal melanoma treated with plaque radiotherapy or helium ion radiotherapy and found neovascular glaucoma in 11% of the plaque-irradiated group and in 29% of the helium ion-

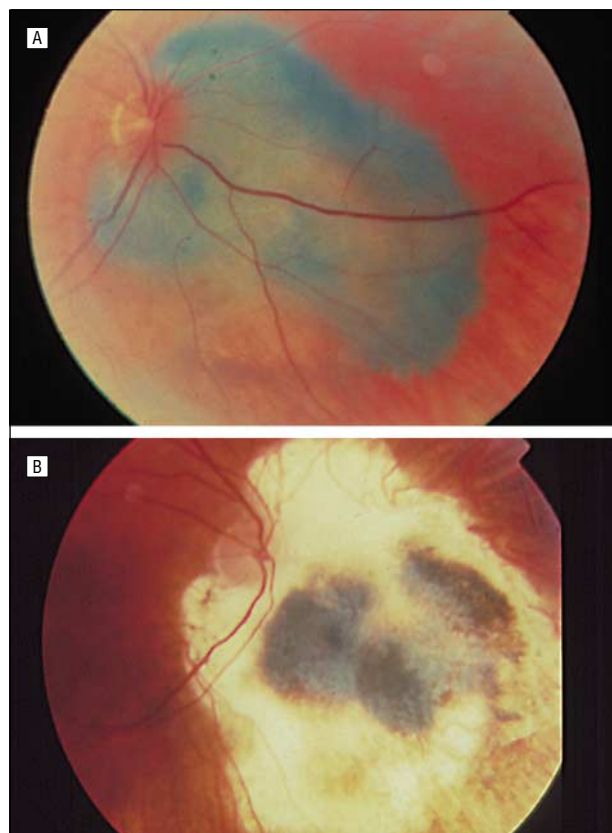


Figure 3. Documented enlarging juxtapapillary choroidal melanoma. A, Before treatment, the pigmented tumor abuts the optic disc and displays orange pigment and associated subretinal fluid. B, Following plaque radiotherapy combined with thermotherapy, the tumor completely regressed and, at 3 years' follow-up, the visual acuity remains 20/25.

treated group. Wilson and Hungerford¹¹ found that severe radiation complications and enucleation were less following plaque radiotherapy compared with proton beam radiotherapy for choroidal melanoma. Eleven (6%) of 190 eyes treated with ¹²⁵I plaque radiotherapy required enucleation compared with 29 (11%) of 267 eyes treated with proton beam radiotherapy. In our group of 270 patients treated with combined plaque radiotherapy and thermotherapy, there were 3 patients (1%) who developed neovascular glaucoma. Using Kaplan-Meier estimates, 7% developed neovascular glaucoma by 5 years' follow-up. Enucleation for radiation complications was performed in 3 cases (1%). This may be partly explained by the fact that the tumors in our series may be slightly smaller with less subretinal fluid than those treated with charged particle radiotherapy. However, in this series we did treat large choroidal melanoma with tumor thickness up to 9 mm.

There are limitations that should be realized in this series of patients. This article provides short-term 6-year results and we expect more local recurrences with longer follow-up. First, as with any report on posterior uveal melanoma, tumor recurrence and distant metastasis are ideally evaluated over a follow-up period of 10 to 20 years. Second, thermotherapy is generally reserved as a supplemental treatment to radiotherapy for choroidal melanoma posteriorly in the eye where it is feasible to provide the heat through the dilated pupil. It may be infeasible

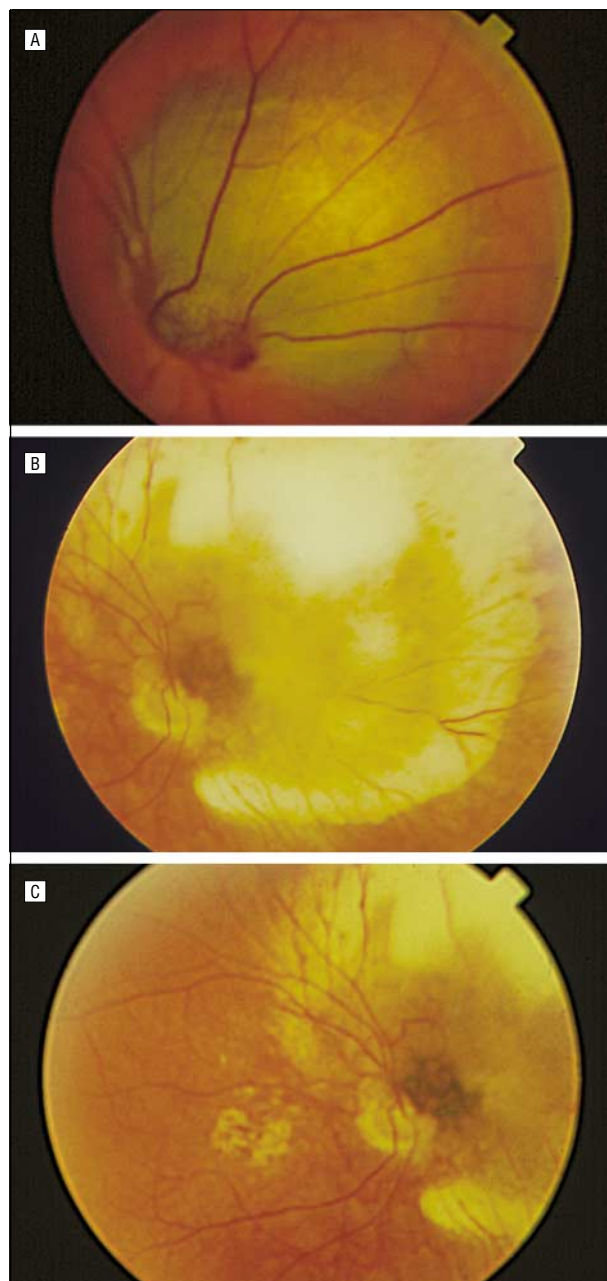


Figure 4. Juxtapapillary choroidal melanoma overhanging the optic disc. A, Before treatment, the pigmented tumor overhanging the optic disc and measured 4.9 mm in thickness. B, Seven years following plaque radiotherapy combined with thermotherapy, the tumor completely regressed to only 1.1-mm thickness. C, At 7 years' follow-up, the visual acuity decreased to hand motions from radiation papillopathy and maculopathy.

to provide thermotherapy to large melanomas, especially those in the peripheral fundus or with substantial retinal detachment. This knowledge could skew our patient selection toward those with smaller tumors, macular or juxtapapillary location, and less subretinal fluid, which could bias results toward a decreased metastatic rate, but possibly increased macular and papillary complications. Third, it is difficult to compare our results with other reports as there are different tumor parameters, treatment techniques, definitions of recurrence, and follow-up intervals. Additionally, some authors have used thermotherapy immediately following plaque radio-

therapy while others have delayed thermotherapy and applied it for tumor nonresponse or tumor recurrence on follow-up.^{14,27} However, it is critical to realize that we represent a tertiary referral center for ocular tumors so that more difficult cases might be managed by us and included in a series such as this. Lastly, the number of tumor recurrences was small, thus the univariable risks predictive for this event are of low power.

In summary, plaque radiotherapy combined with transpupillary thermotherapy provides excellent intraocular tumor control with local tumor recurrence of only 3% at 5 years' follow-up. Hopefully, such control will reflect long-term improved life prognosis.

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